

Matrix inversion lemmas

by Marco Taboga, PhD

Matrix inversion lemmas are extremely useful formulae that allow us to efficiently compute how simple changes in a matrix affect its [inverse](#).



Rank one updates

One of the simplest changes that can be performed on a matrix is a so-called rank one update.

Definition Let A be a $n \times n$ matrix and u and v two $n \times 1$ column vectors. Then, the transformation

$A \rightarrow A + uv^T$ is called a rank one update to A .

The reason why the transformation is called rank one is that the [rank](#) of the matrix $A + uv^T$ is equal to $\text{rank}(A) + 1$ (because a single vector, u , [spans](#) all the columns of uv^T).

Rank one updates to the identity matrix

The first inversion lemma we present is for rank one updates to identity matrices.

Proposition Let I_n be the $n \times n$ identity matrix and u and v two $n \times 1$ column vectors. The matrix

$I_n + uv^T$ is invertible if and only if

When it is invertible, its inverse is

$$(I + uv^T)^{-1} = I - \frac{1}{1 + v^T u} uv^T$$

Proof

Note the dimensions of the **matrix products** involved in this inversion lemma:

- the rank one update, that is, the product

is ;

- the invertibility condition is based on the product

whose dimension is ; in other words, it is a scalar.

Sherman-Morrison formula

The Sherman-Morrison formula, shown in the next proposition, generalizes the previous inversion lemma by considering rank one updates to a generic invertible matrix (instead of only considering rank one updates to the identity matrix).

Proposition Let be a invertible matrix and and two column vectors. The rank one update

is invertible if and only if

$$1 + v^T A^{-1} u \neq 0$$

When it is invertible, its inverse is

$$(A + uv^T)^{-1} = A^{-1} - \frac{1}{1 + v^T A^{-1} u} A^{-1} uv^T A^{-1}$$

Proof

The Sherman-Morrison formula is very useful not only because rank one updates are found in many theorems in matrix algebra, but also because it can be a computationally efficient way of calculating the inverse when A^{-1} has already been calculated. The number of arithmetic operations needed to compute A^{-1} from scratch is proportional to n^3 , while the number of operations needed to update the inverse with Sherman-Morrison formula is proportional to n^2 . Even if in the latter case the constant of proportionality is higher than in the former, when n is sufficiently large, the computational cost of the update is much lower.

Woodbury matrix identity

Another useful matrix inversion lemma goes under the name of Woodbury matrix identity, which is presented in the following proposition.

Proposition Let A be a $n \times n$ invertible matrix, and U and V two $n \times m$ matrices, and C an $m \times m$ invertible matrix. If

$A + UCV^T$ is invertible, then

$C + V^T A^{-1} U$ is invertible and its inverse is

$$(A + UCV^T)^{-1} = A^{-1} - A^{-1}U(C^{-1} + V^T A^{-1}U)^{-1}V^T A^{-1}$$

Proof

Note that when $m = 1$ and $C = 1$, the Woodbury matrix identity coincides with the Sherman Morrison formula. Therefore, the latter is a special case of the former.

The reasons why this inversion lemma is worth knowing are similar to those we have explained for the Sherman Morrison formula: it is often used in matrix algebra, and it saves computations when A^{-1} is already known (and m is significantly smaller than n).

How to cite

Please cite as:

Taboga, Marco (2021). "Matrix inversion lemmas", Lectures on matrix algebra.

<https://www.statlect.com/matrix-algebra/matrix-inversion-lemmas>.



The books

Most of the learning materials found on this website are now available in a traditional textbook format.

Probability and statistics

Matrix algebra

Featured pages

Score test
Beta function
Law of Large Numbers
Convergence in distribution
Moment generating function
Permutations

Main sections

Mathematical tools
Fundamentals of probability
Probability distributions
Asymptotic theory
Fundamentals of statistics
Glossary

Explore

Student t distribution
Central Limit Theorem
F distribution

About

About Statlect
Contacts
Cookies, privacy and terms of use

Glossary entries

Critical value
Continuous mapping theorem
Binomial coefficient
Distribution function
Null hypothesis
IID sequence

Share

To enhance your privacy,
we removed the social buttons,
but **don't forget to share**.